Multiple Traffic Light Controller with Remote IoT Access

Final Project

Francisco Acosta | CEIS 114 | Feb 2024

Introduction

The following presentation covers my journey of creating a fully integrated multiple traffic light controller system with a crosswalk button, buzzer, and an emergency light with remote IoT access.

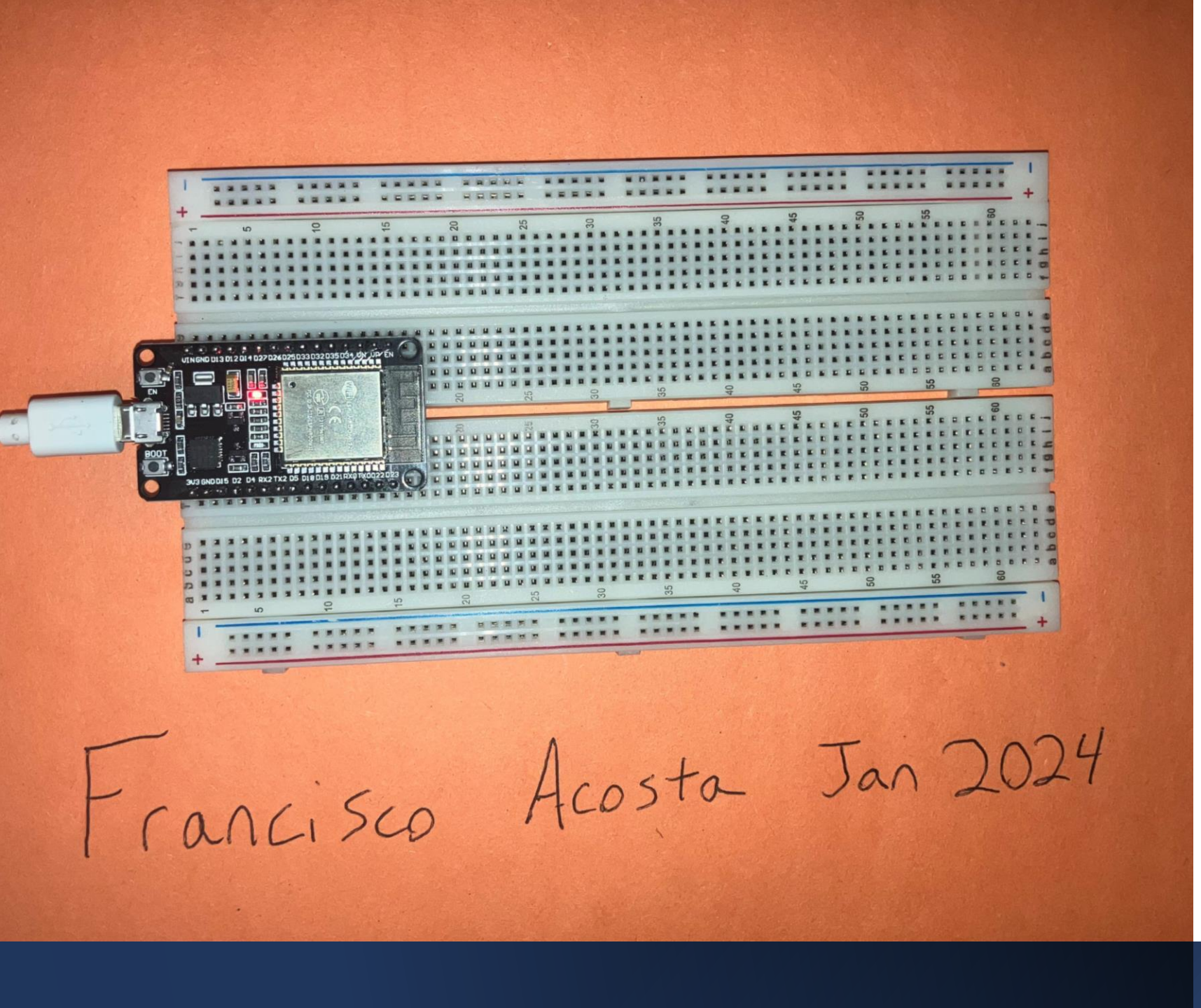
The use of circuitry, networking, digital components, and integrated development environments will be displayed throughout the course of this build.

Please follow along to explore the real-world simulation of an IoT device.

Inventory

- ESP32 Microcontroller
- Two sets of colored LEDs: Red, Yellow, and Green
- One Blue LED
- Wires
- Two Breadboards
- LCD Unit
- Active Buzzer
- Mini Router
- Push Buttons
- PIR Motion Sensor

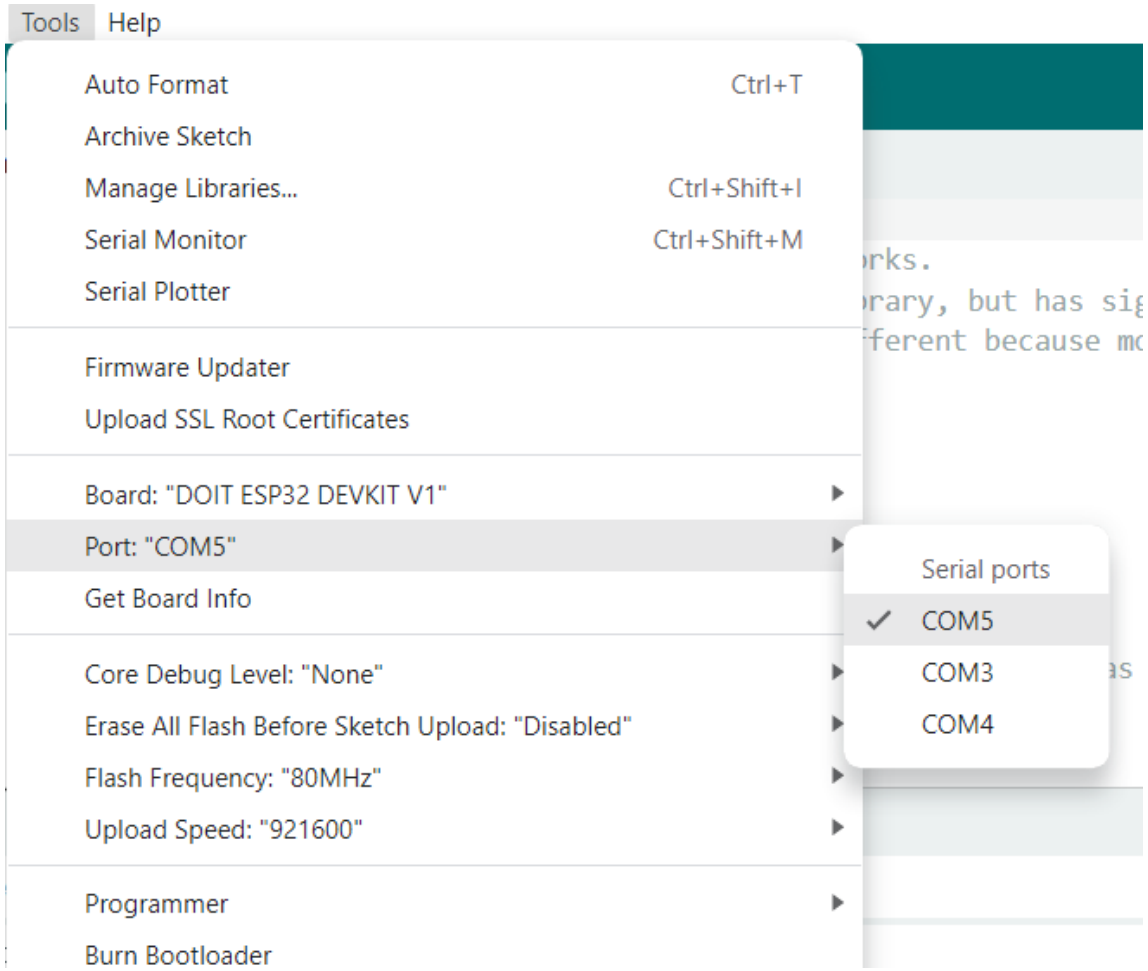




ESP32 Microcontroller Mounted

The initial phase of this build started with the mounting of an ESP32 microcontroller to a pair of breadboards.

Installation of Arduino IDE



Arduino was the platform of choice, and a fresh install was required for my computer.

This was followed by connecting the device to my machine and identify which port was being utilized.

ESP32 Wi-Fi Scan

Output Serial Monitor ×

Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM5')

```
Scan done
5 networks found
Nr | SSID                     | RSSI | CH | Encryption
 1 | ATTYGwSfYs               |  -76 |  1 | WPA2
 2 | SpectrumSetup-70         |  -80 |  6 | WPA2
 3 | SpectrumSetup-76         |  -86 |  6 | WPA2
 4 | ATT7W4y7Y7               |  -89 |  1 | WPA2
 5 | Sosa's Home              |  -91 |  2 | WPA2

Scan start
Scan done
4 networks found
Nr | SSID                     | RSSI | CH | Encryption
 1 | ATTYGwSfYs               |  -79 |  1 | WPA2
 2 | SpectrumSetup-70         |  -80 |  6 | WPA2
 3 | SpectrumSetup-76         |  -85 |  6 | WPA2
 4 | ATT7W4y7Y7               |  -89 |  1 | WPA2
```

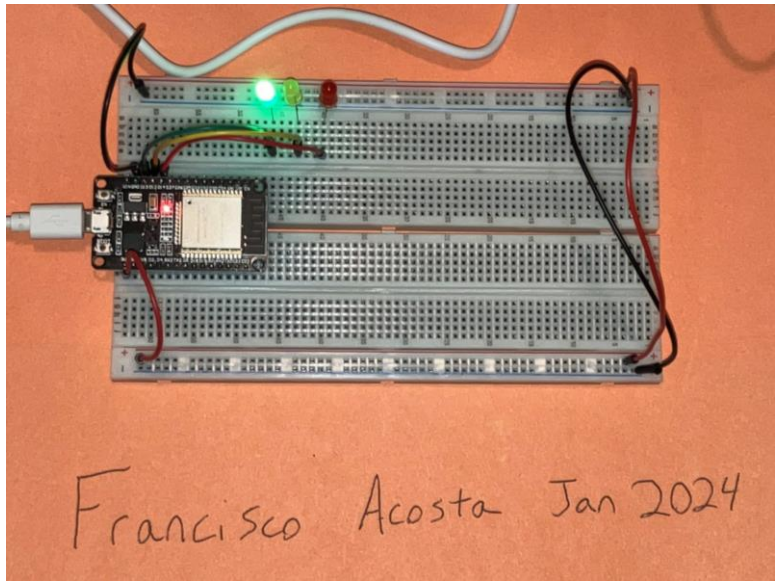
A Wi-Fi scan was necessary to enable access to the internet and proceed with the remainder of the project build.

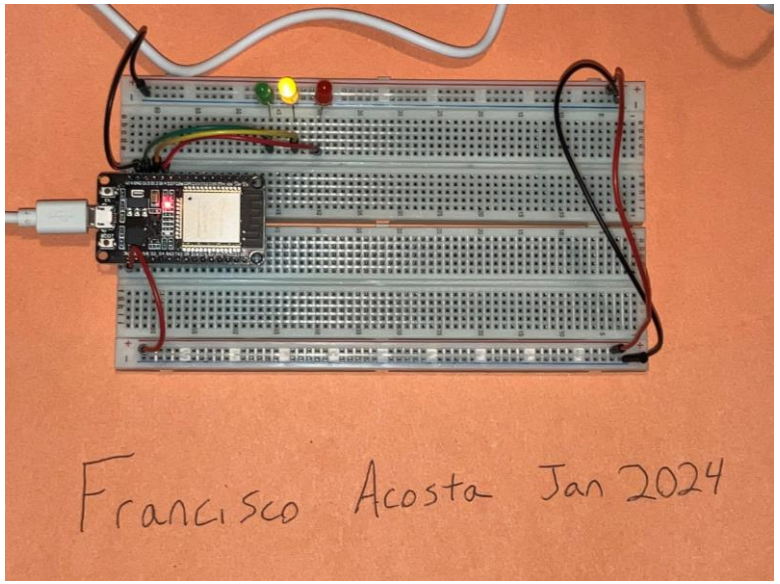
Traffic Lights

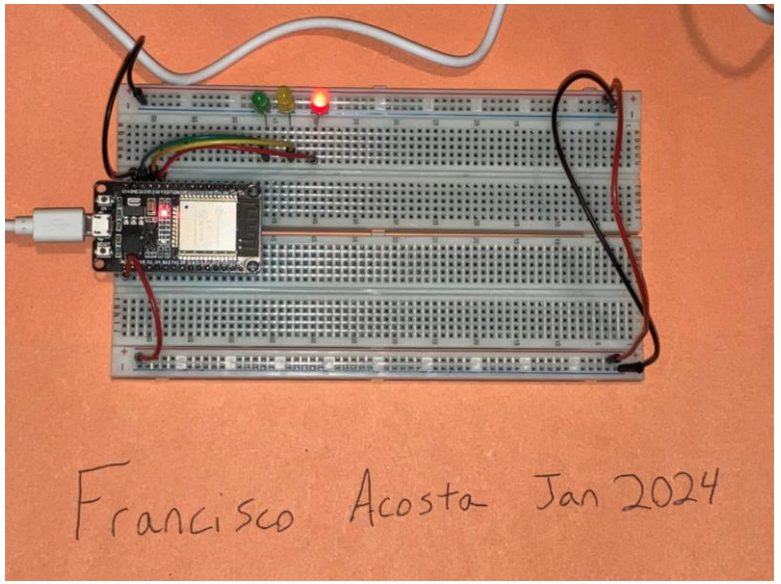
Red, yellow, and green LED lights were connected to the ESP32 microcontroller and breadboard to signify a traffic light you would traditionally see on the roadway.

Code was then created and uploaded to the ESP32 microcontroller to test if each LED would correctly illuminate.

Circuit with working LEDs







Arduino code for LED testing

File Edit Sketch Tools Help

Select Board

sketch_jan14a.ino

```
// Francisco Acosta
// CEIS 114 - Module #3 Project Jan 2024

const int red_LED1  = 14;       // The red LED1 is wired to ESP32 board pin GPIO14
const int yellow_LED1  = 12;     // The yellow LED1 is wired to ESP32 board pin GPIO1
const int green_LED1 = 13;  // The green LED1 is wired to ESP32 board pin GPIO13

// the setup function runs once when you press reset or power the board
void setup() {
 pinMode(red_LED1, OUTPUT);     // initialize digital pin GPIO14 (Red LED1) as an out
pinMode(yellow_LED1, OUTPUT);   // initialize digital pin GPIO12 (yellow LED1) as an
pinMode(green_LED1, OUTPUT);      // initialize digital pin GPIO13 (green LED1) as an
}

// the loop function runs over and over again forever
void loop() {
  // The next three lines of code turn on the red LED1
  digitalWrite(red_LED1, HIGH);       // This should turn on the RED LED1
  digitalWrite(yellow_LED1 , LOW);    //  This should turn off the YELLOW LED1
  digitalWrite(green_LED1, LOW);      //  This should turn off the GREEN LED1

  delay(2000);                       // wait for 2 seconds

  // The next three lines of code turn on the green LED1
  digitalWrite(red_LED1, LOW);         // This should turn off the RED LED1
  digitalWrite(yellow_LED1 , LOW);        // This should turn off the YELLOW LED1
  digitalWrite(green_LED1, HIGH);        // This should turn on the GREEN LED1

  delay(2000);                     // wait for 2 seconds

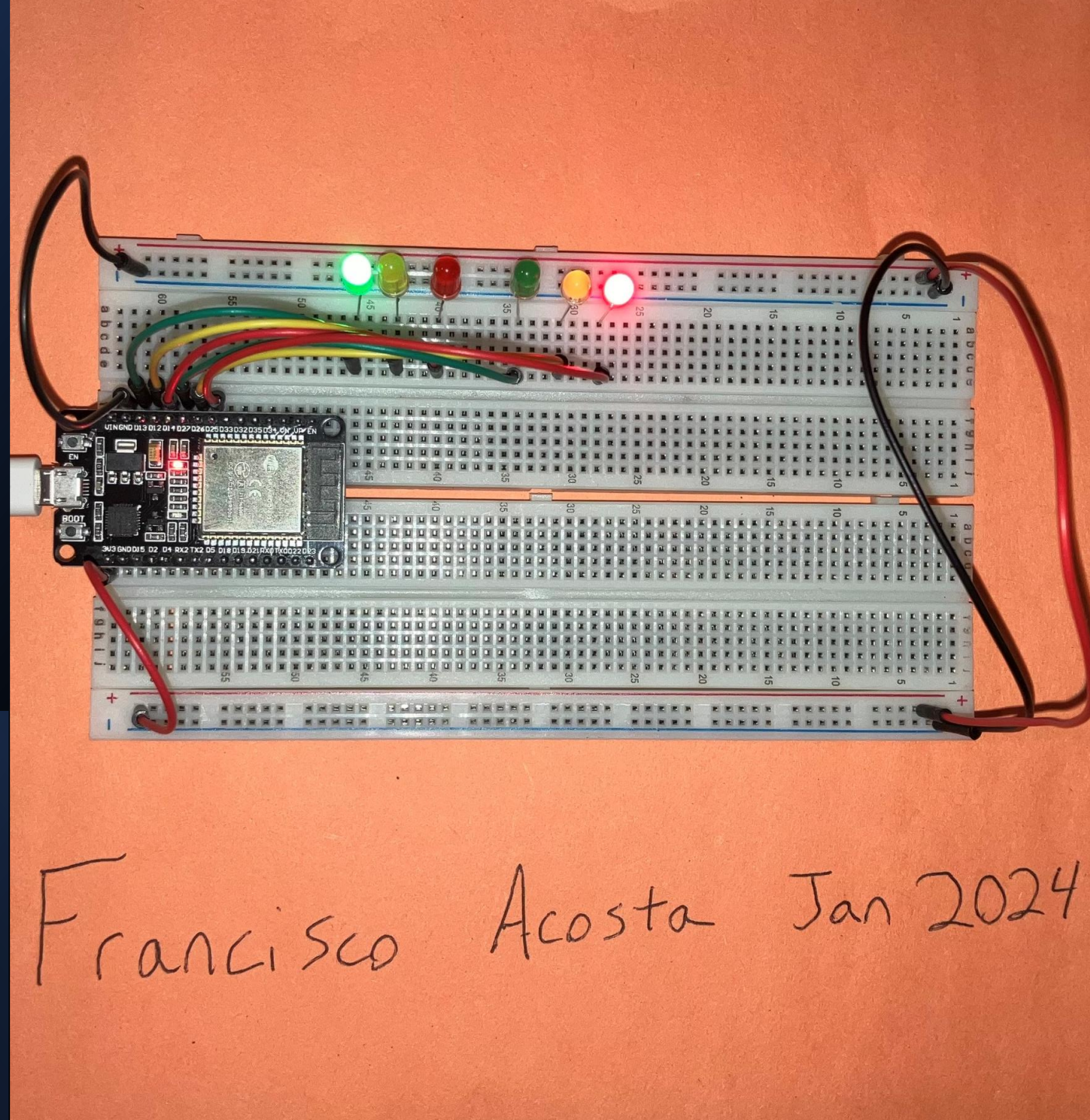
// The next three lines of code turn on the yellow  LED1
   digitalWrite(red_LED1, LOW);       // This should turn off the RED LED1
  digitalWrite(yellow_LED1 , HIGH);   //  This should turn on the YELLOW LED1
  digitalWrite(green_LED1, LOW);        //  This should turn off the GREEN LED1

  delay(2000);                       // wait for 2 seconds
}
```

Second Set of Traffic Lights

An additional set of lights were implemented to signify a second traffic light that simulates an intersection.
The circuit accounts for standard traffic regulations allowing all LED lights to illuminate accordingly.
One traffic light will activate a red LED while the other cycles through red, yellow and green.
The traffic lights will alternate once the first cycle is complete.

Circuit with a pair of working LEDs



Arduino code adding functionality to both sets of LEDs

DOIT ESP32 DEVKIT V1

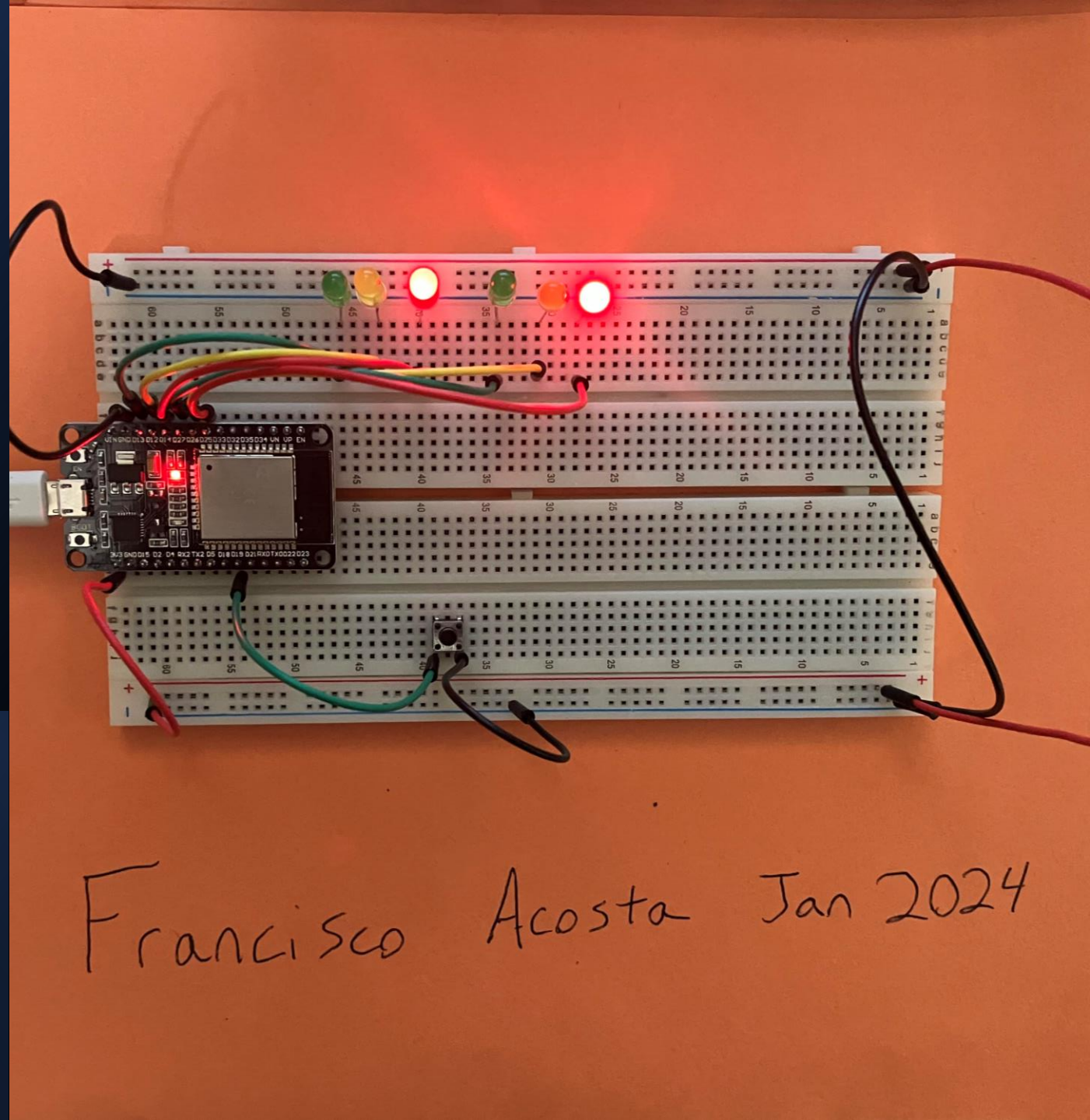
ArduinoProject.ino

```
1   // Francisco Acosta
2   // CEIS 114 - Module #4 Project - Jan 2024
3
4   // Define some labels
5   const int red_LED1  = 14;   // The red LED1 is wired to ESP32 boa
6   const int yellow_LED1  =12;   // The yellow LED1 is wired to ESP3
7   const int green_LED1 = 13; // The green LED1 is wired to ESP32 bo
8   const int red_LED2  = 25;   // The red LED2 is wired to Mega boar
9   const int yellow_LED2  = 26;   // The yellow LED2 is wired to Meg
10  const int green_LED2 = 27; // The green LED2 is wired to Mega boa
11
12  // the setup function runs once when you press reset or power the
13  void setup() {
14    pinMode(red_LED1, OUTPUT);  // initialize digital pin GPIO14 (Re
15    pinMode(yellow_LED1, OUTPUT);  // initialize digital pin GPIO12
16    pinMode(green_LED1, OUTPUT);    // initialize digital pin GPIO13
17    pinMode(red_LED2, OUTPUT);  // initialize digital pin GPIO25(Red
18    pinMode(yellow_LED2, OUTPUT); // initialize digital pin GPIO26 (
19    pinMode(green_LED2, OUTPUT); // initialize digital pin GPIO27 (g
20  }
21
22  // the loop function runs over and over again forever
23  void loop() {
24    // The next three lines of code turn on the red LED1
25    digitalWrite(red_LED1, HIGH);      // This should turn on the R
26    digitalWrite(yellow_LED1 , LOW);        //  This should turn of
27    digitalWrite(green_LED1, LOW);       //  This should turn off
```

Crosswalk

A push button was integrated into the circuit for the purpose of simulating a crosswalk.
Pushing the button activates a sequence of events that will have both traffic lights illuminating their red LED lights after their current cycle is complete.
A count of ten seconds would elapse before the traffic lights resume their standardized functions.

Push button added to the circuit



Arduino code adding functionality to the push button

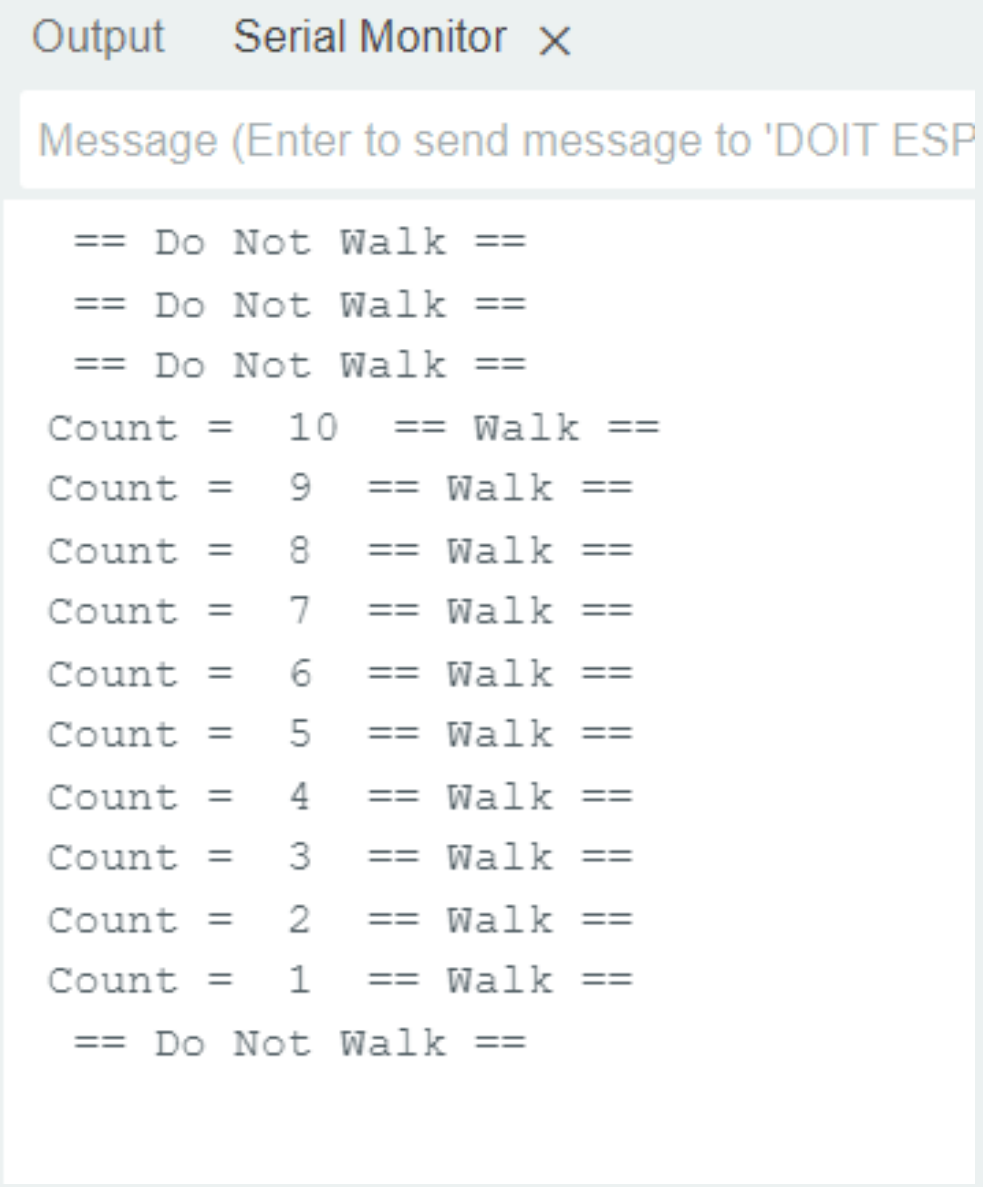
edit Sketch Tools Help

DOIT ESP32 DEVKIT V1

ArduinoProject.ino

```
1  //Francisco Acosta --  CEIS 114  --  Feb 2024
2  // Module #5 Project
3  const int red_LED1  = 14;   // The red LED1 is wired to ESP32 board
4  const int yellow_LED1  =12;   // The yellow LED1 is wired to ESP32 b
5  const int green_LED1 = 13; // The green LED1 is wired to ESP32 board
6  const int red_LED2  = 25;   // The red LED2 is wired to Mega board p
7  const int yellow_LED2  = 26;   // The yellow LED2 is wired to Mega b
8  const int green_LED2 = 27; // The green LED2 is wired to Mega board
9
10 int Xw_value;
11 const int Xw_button = 19; //Cross Walk button
12
13 // the setup function runs once when you press reset or power the bo
14 void setup() {
15
16   pinMode(Xw_button, INPUT_PULLUP); // 0=pressed, 1 = unpressed butto
17   Serial.begin(115200);
18   pinMode(red_LED1, OUTPUT);  // initialize digital pin 14 (Red LED1)
19   pinMode(yellow_LED1, OUTPUT);  // initialize digital pin 12 (yellow
20   pinMode(green_LED1, OUTPUT);    // initialize digital pin 13 (green
21
22   pinMode(red_LED2, OUTPUT);  // initialize digital pin 25(Red LED2)
23   pinMode(yellow_LED2, OUTPUT); // initialize digital pin 26 (yellow
24   pinMode(green_LED2, OUTPUT); // initialize digital pin 27 (green LE
25 }
```

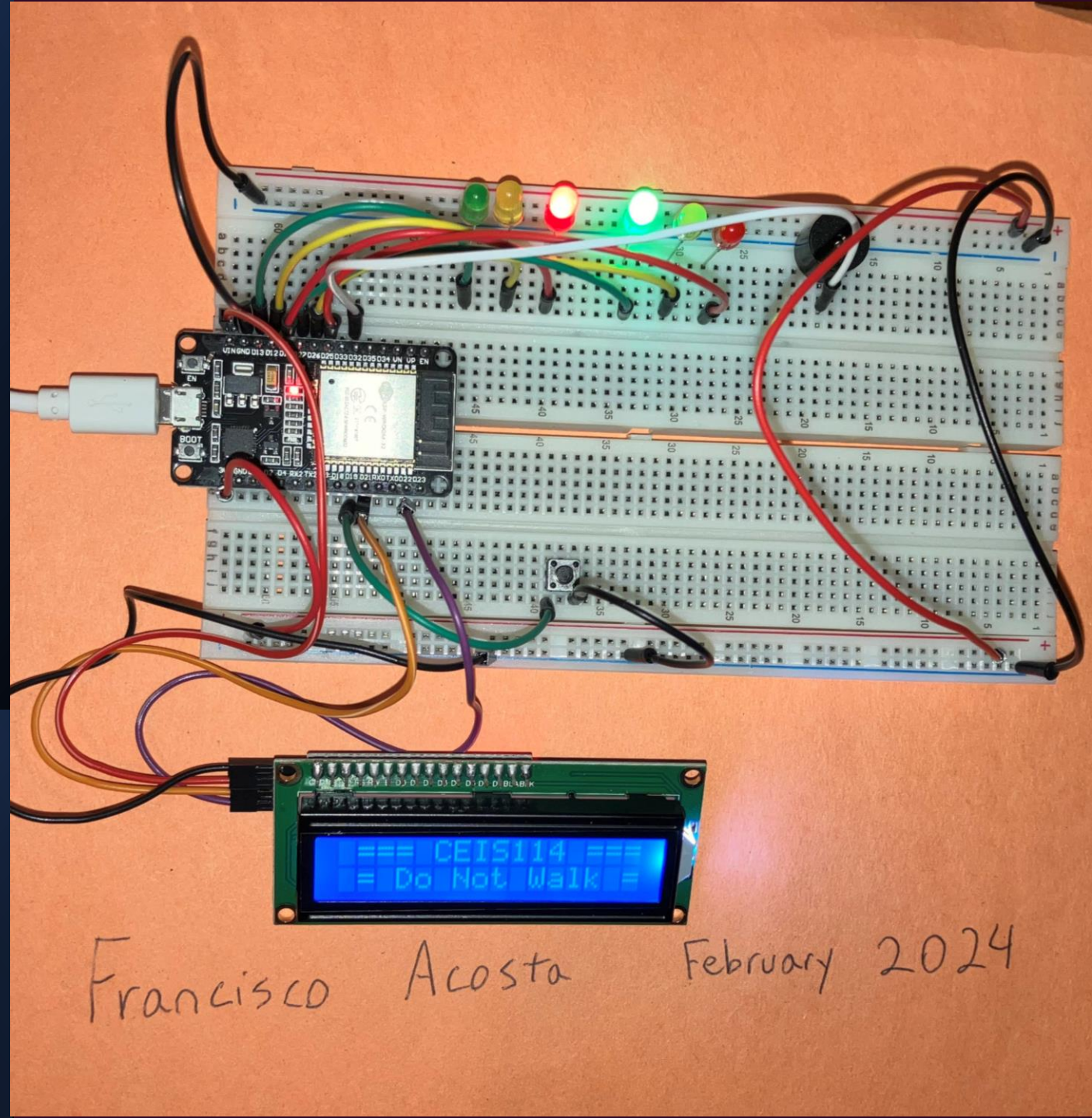
Arduino serial monitor displaying the correct output of a functional push button



Buzzer and LCD

A buzzer was added to radiate a sound and an LCD unit was added to display a message.
They are both utilized to alert pedestrians when it is safe to walk across the intersection.

Circuit with the buzzer and LCD unit added



Arduino code with the added functionality for the LCD unit and buzzer

DOIT ESP32 DEVKIT V1

ArduinoProject.ino

```
1   // === Francisco Acosta ====
2   // Module #6 Project  February 2024
3   #include <Wire.h>  //lcd
4   #include <LiquidCrystal_I2C.h> //lcd
5   LiquidCrystal_I2C lcd(0x27,16,2); //set the LCD add
6   // if it does not work then try 0x3F, if both addre
7
8   const int bzr=32;      // GPIO32 to connect the Buz
9   //=================== LCD ===================
10  const int red_LED1  = 14;   // The red LED1 is wire
11  const int yellow_LED1  =12;   // The yellow LED1 is
12  const int green_LED1 = 13; // The green LED1 is wir
13  const int red_LED2  = 25;   // The red LED2 is wire
14  const int yellow_LED2  = 26;   // The yellow LED2 i
15  const int green_LED2 = 27; // The green LED2 is wir
16
17  int Xw_value;
18  const int Xw_button = 19; //Cross Walk button
19
20  void setup() {
21    Serial.begin(115200);
```

Arduino serial monitor outputting the count utilized for pedestrian crossing

Output Serial Monitor ×

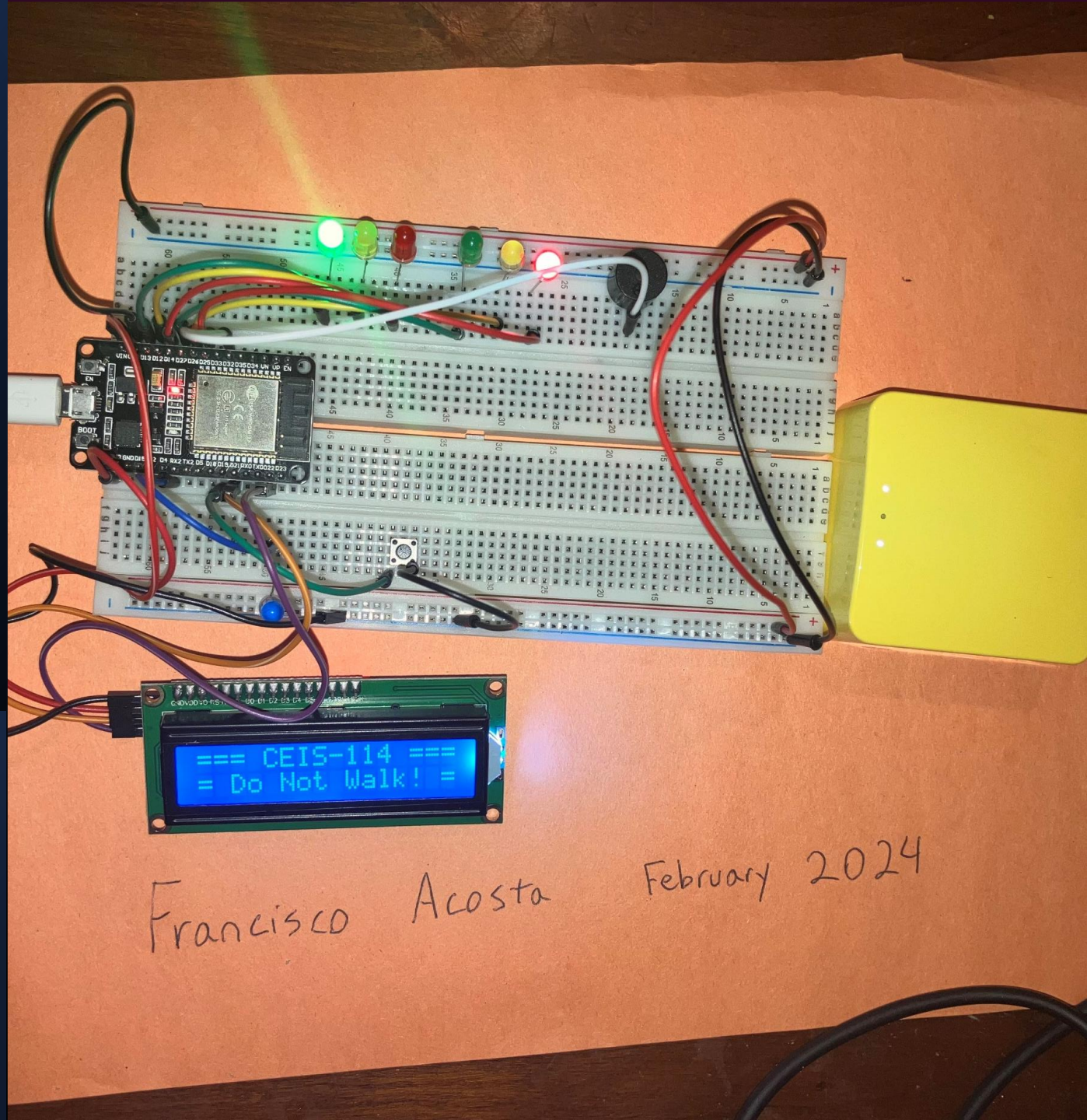
Message (Enter to send message to '

```
 == Do Not Walk ==
 == Do Not Walk ==
 == Do Not Walk ==
 == Do Not Walk ==
 == Do Not Walk ==
Count =  10  == Walk ==
Count =  9  == Walk ==
Count =  8  == Walk ==
Count =  7  == Walk ==
Count =  6  == Walk ==
Count =  5  == Walk ==
Count =  4  == Walk ==
Count =  3  == Walk ==
Count =  2  == Walk ==
Count =  1  == Walk ==
Count =  0  == Walk ==
 == Do Not Walk ==
```

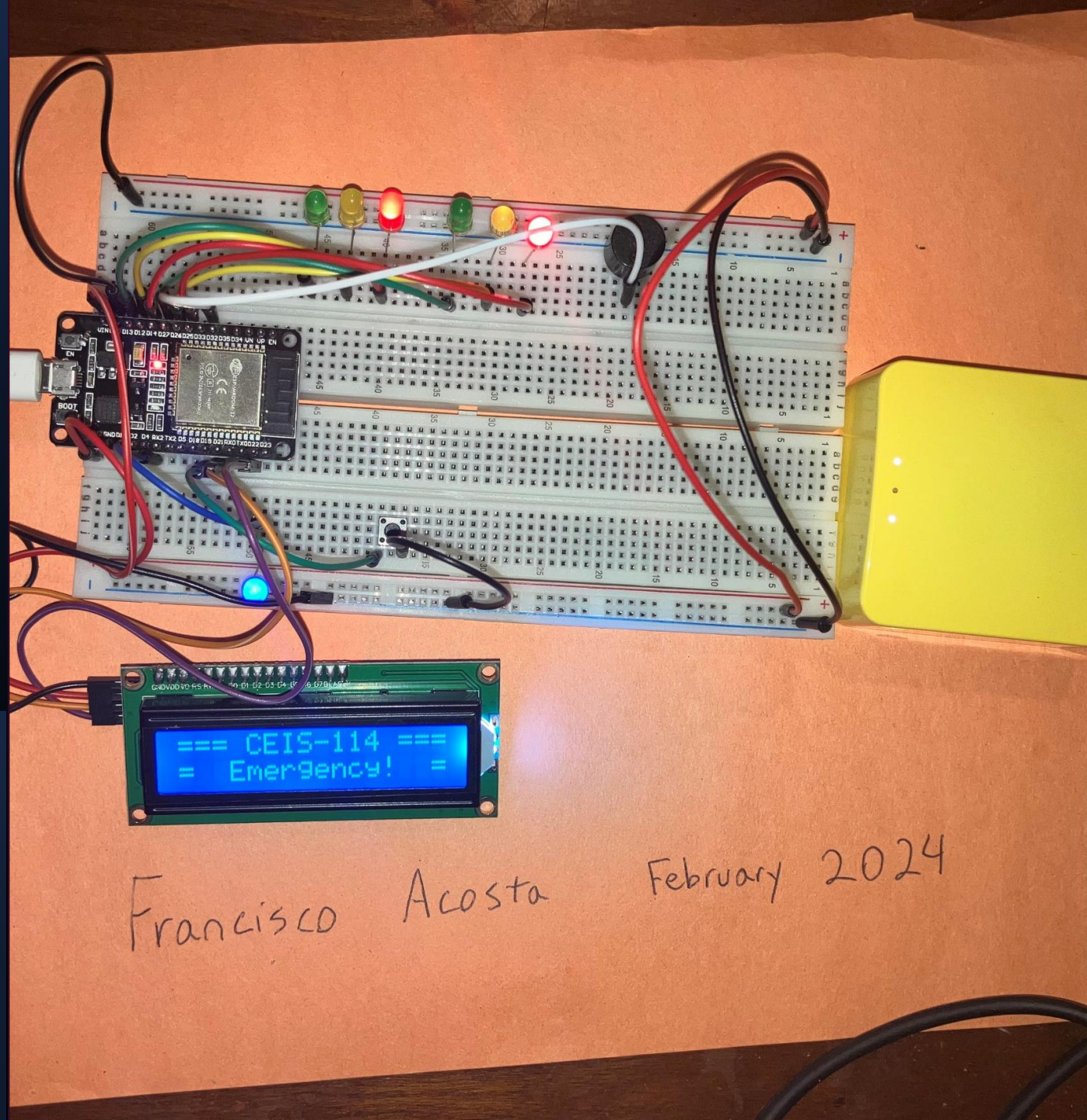
Remote IoT Access

The next stage adds a mini router and Blue LED light to the device. The Blue LED represents an emergency signal that halts all traffic and notifies those around of an emergent situation. First responders would be given the ability to halt all traffic by illuminating this blue LED. The mini router enables internet access and an external application such as IoT MQTT Panel can turn on and off the blue LED.

LCD with the message “Do Not Walk!” and the blue LED off



LCD with the message “Emergency!” and the blue LED on

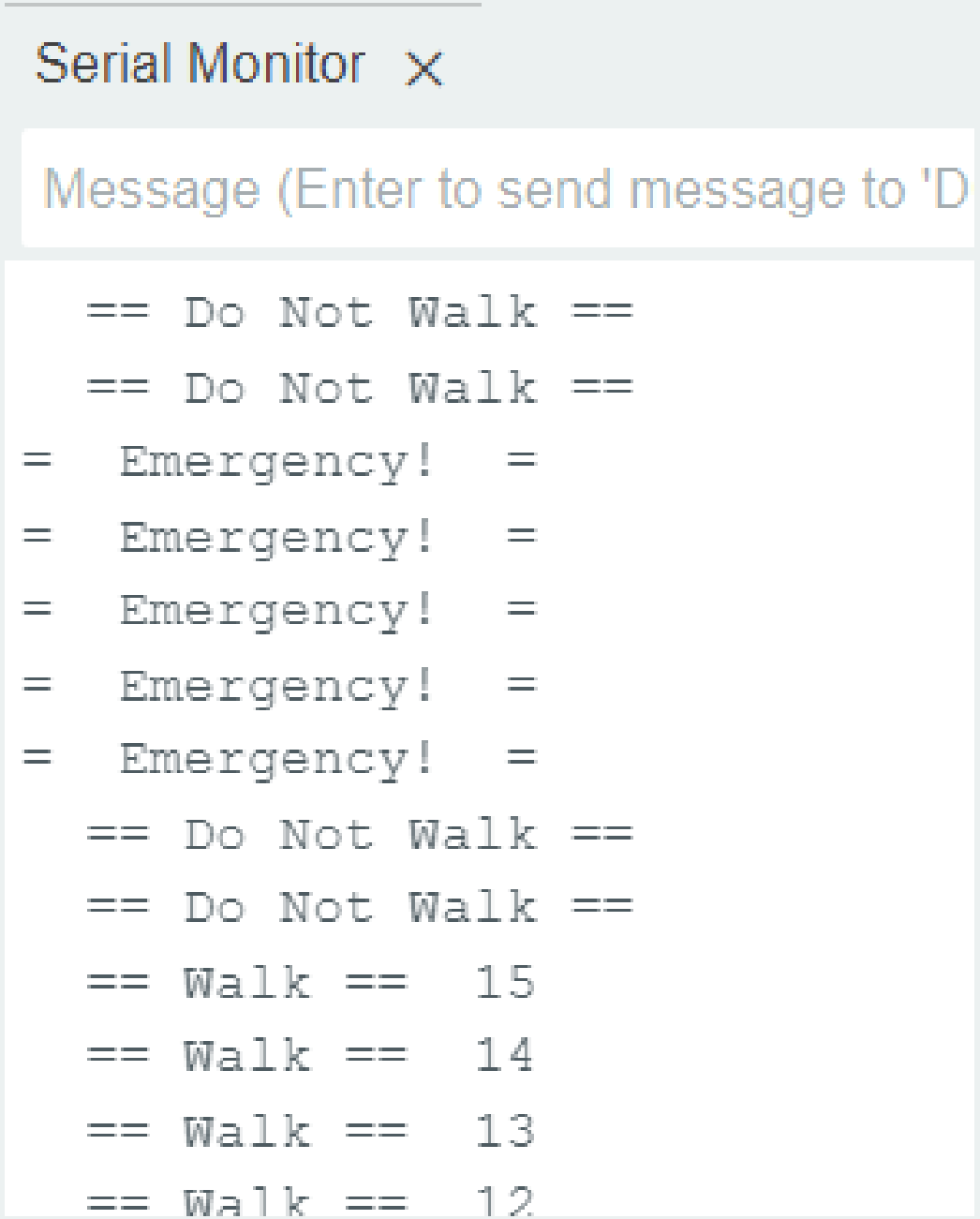


Arduino code implementing remote IoT access

ArduinoProject.ino

```
1   // === Francisco Acosta ====
2   // Final Project Component, Option 1, February 2024
3
4   #include <WiFi.h>      // WiFi header file
5   #include <PubSubClient.h>    // MQTT publish and subscribe header file
6   #include <Wire.h>                              // I2C header file
7   #include <LiquidCrystal_I2C.h>                 // I2C lcd header file
8
9
10  const char* ssid       = "GL-MT300N-V2-8e1";  // This is the access point
11                         // Please check the guide at the bottom of the project guide
12                         // Replace the xxxxxxxxx characters with the rest f the name
13
14  const char* password   = "goodlife";  // This is the password to the SSID
15                                                                  // The pas
16  const char* mqttServer = "test.mosquitto.org";  // This is the free MQTT
17  int port = 1883;   // MQTT brokers listen to port 1883 by default
18  String stMac;      // C string used for convenience of comparisons.
19  char mac[50];     // C char array used to hold the MAC address of your ES
20  char clientId[50];   // This client ID is used to identify the user acces
```

Arduino serial monitor displaying a successful remote IoT connection



Teamwork

The ability to work within a team is crucial.
I was able to share the planning stage of this build with my peers and read over their plans. Everyone was given an opportunity to share their feedback on each other's strategy of completing their final presentation. The end of this module allowed me to reflect on their responses and share my thoughts.

Teamwork

Planning



Francisco Acosta

Monday

I chose to build option 1, which is a traffic light device with the integration of IoT control using cloud services.
Option 1 seems to suit me best because cloud computing is a fascinating topic that genuinely interests me.
I also picked this option because it will aid me in the transition to my next course “Fundamentals of Cloud Computing”.
Any steps to further my education is vital to my success, and this is an opportunity to capitalize on an important experience.

My preparation includes the completion of the IoT device in its entirety to eliminate all tasks outside of the PowerPoint presentation.
The goal is to focus all my attention on the presentation portion of the project.
I will then plan to import all prior PowerPoint projects into one master file and organize the project slides sequentially.
The next process would be the consolidation of slides to eliminate redundancy.
I would then proofread and make improvements as I see fit.
The last item would be to do a final review of the project as a whole to see if the slideshow meets all requirements and is professionally presentable.

Reply (1 like)

Teamwork

Response to Feedback



Francisco Acosta

3:11pm

Jose,
I am a firm believer that strategically planning in a methodically organized manner leads to success.
Proactive thinking prepares an individual for most scenarios and gaining clarity allows for complete focus of the task at hand.
Attention to detail is a key component to pushing a project to the next level and creates an effective delivery.
I appreciate the generous words of encouragement you provided and will keep all of your feedback in mind as I progress through my final presentation.

Reply

Teamwork

Feedback to Peer



Stephen Swiergosz

Tuesday

Option 2

After some consideration, I have decided to work on Option 2. This option, which involves designing traffic lights with continuous activation in one direction of the major street, integrating a motion detector for the minor street, and adding a push button for an emergency signal,

I chose Option 2 because real-world Connection: It closely relates to real traffic situations, making our project more practical.

To prepare for the Final Project Deliverable, I will start by thoroughly reviewing the project guide document and watching the associated video series. I plan to follow the instructions closely, making necessary adjustments and alterations to previous PowerPoints I created. Including key slides or information from those presentations

Reply



Francisco Acosta

Tuesday

Stephen,
Utilizing the provided documentation as a reference guide while piecing together your presentation is a solid plan.
Consistently reviewing and making alterations throughout the project build is a great strategy.
I would suggest laying out a basic structure of your own within PowerPoint.
This will allow you to easily add and revise content as you progress through your final presentation.
Focusing on the content necessary for each section of the project will therefore become the main priority.
This would be in addition to your current strategy of following the provided instructional videos and documentation specified for this project. I believe you will effectively communicate your project with ease and I wish you the best of luck in your future endeavors.

Reply

Teamwork

Feedback to Peer



Jose Manjarres (*He/Him*)

Yesterday

I've made the decision to focus on Option 2 for our project. This option involves implementing traffic lights with continuous activation on one side of the main street, incorporating a motion sensor for the side street, and integrating a push button for emergency signals.

I've opted for Option 2 because of its strong connection to real-world traffic scenarios, which adds practicality to our project.

To prepare for the Final Project Deliverable, I'll begin by thoroughly reviewing the project guide document and going through the associated video series. My plan is to closely adhere to the instructions, making any necessary adjustments or modifications to the PowerPoints I've previously prepared. This may involve incorporating important slides or information from those presentations.

Reply



Francisco Acosta

Yesterday

Jose,
Your strategy to lean on previous material while modifying content to improve quality is fundamentally solid.
I would recommend the consolidation of presentation slides as you combine previous projects to eliminate redundancy.
The reliance on the outline provided is great way to stay on the most successful path possible.
I believe you will effectively complete this project and your efforts will lead you down the path to success.
I wish you well on your journey and anticipate a bright future ahead.

Reply

Challenges

Finding the proper driver for the ESP32 microcontroller was one of the first challenges I encountered. I was able to finally find the ESP32 CP210x USB to UART Bridge VCP Drivers after searching for it online with the provided documentation.

Another challenge I came across was finding the correct Liquid Crystal information within Arduino to properly display a message. I eventually found the solution by modifying the provided code.

I also struggled creating a successful IoT connection from the MQTT client to the traffic controller but was able to resolve this by ensuring all settings were correctly enabled.

The final challenge was the biggest. Compiling all modules into a clear and concise presentation was certainly a feat and took a bit of time to create a polished final product.

Obstacles are inevitable and this project was no exception. I find learning to overcome these challenges and finding a way to succeed is tremendously rewarding.

Career Skills

- I gained valuable experience utilizing digital components consisting of LEDs, an ESP32 microcontroller, an LCD, and the circuitry connecting them together.
- I was able to apply the principles of networking by providing remote IoT access to the traffic controller utilizing a mini router and a MQTT client.
- I was able to gain working experience within an integrated development environment utilizing Arduino.
- The most treasured career skill gained was real-world experience developing a traffic controller.

Conclusion

This was an insightful introduction to digital devices that enabled me to gain valued real-world experience creating a traffic controller with remote IoT access. Creating a fully integrated system opened the door to circuitry, networking, digital components and integrated development environments with an introduction to small amounts of programming. The conclusion of this course provided me with the knowledge and tools necessary to begin my journey of joining the workforce as a future contributor to the technology industry.

Reference

Arduino. 2024.
https://www.arduino.cc/

Silicon Laboratories. 2024.
https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers